Game Theory for Data Science

Problem Set Exam (take-home final) • Fall 2023

Instructor: Dr. S. Iyer • 100 points

Released Fri Dec 8, 5:00 pm • Due Sun Dec 10, 5:00 pm

Rules for this exam.

- You have 48 hours. The exam is designed to take about 4 hours of focused work.
- Open book and open notes: you may use the course notes, the textbooks, and your own past problem sets. You may *not* discuss the exam with anyone (including on forums) or use AI assistants. Clarification questions go to the instructor by private message only.
- You may use Python (or any language) for arithmetic, but every answer must be derived and explained in words and equations. If you use code, append it; code alone earns no credit.
- Submit one PDF to the course site. Typeset or clearly scanned handwriting are both fine. Give exact fractions where possible.

Problem	1	2	3	4	5	Total
Topic	Correlated eq.	Hedge	Fictitious play	Congestion	True/False	
Points	20	20	20	20	20	100

Problem 1: Drones at a corridor (20 points)

Two delivery drones approach a narrow corridor from opposite ends. Each chooses to *Dare* (D, fly through) or *Chicken* (C, yield and wait). Payoffs (row, column):

	D	C
D	0, 0	8, 3
C	3, 8	6, 6

(a) Find all Nash equilibria (pure and mixed) and each player's expected payoff in each. (5 pts)

(b) A traffic controller draws a profile from a distribution $z = (z_{DD}, z_{DC}, z_{CD}, z_{CC})$ and privately tells each drone its own component. Write the four incentive (obedience) constraints that make z a correlated equilibrium, and simplify each to the form "(linear combination of z's) ≥ 0". (5 pts)

(c) Solve the linear program that maximizes the sum of expected payoffs over all correlated equilibria. Report the optimal z and each player's payoff, and explain in one or two sentences why the controller cannot put more weight on (C, C). (6 pts)

(d) A simpler controller puts probability $1/3$ on each of (D, C), (C, D), (C, C). Is this a correlated equilibrium? Compare the players' payoffs with those in (a) and (c). (4 pts)

Problem 2: Hedging over forecasters (20 points)

A monitoring system combines three forecasters (experts) that each predict whether a server will overload in the next hour. Each hour, forecaster i incurs loss $\ell_t(i) = 1$ if its prediction is wrong and 0 otherwise. The system runs Hedge with multiplicative update $w_{t+1}(i) = w_t(i) \cdot \beta^{\ell_t(i)}$ with $\beta = \frac{1}{2}$ (that is, $\eta = \ln 2$), starting from $w_1 = (1, 1, 1)$, and in round t follows forecaster i with probability $p_t(i) = w_t(i) / \sum_j w_t(j)$. Its expected loss in round t is $\sum_i p_t(i)\ell_t(i)$.

t	losses ℓ_t	weights w_t	probabilities p_t	exp. loss
1	(1, 0, 1)	(1, 1, 1)		
2	(0, 1, 1)			
3	(1, 1, 0)			
4	(0, 1, 0)			

(a) Complete the table (copy it into your submission). What is the algorithm's total expected loss over the four rounds? (8 pts)

(b) Compute the algorithm's regret with respect to the best single forecaster in hindsight. Compare it with the guarantee $\text{Regret}_T \le (\ln n)/\eta + \eta T/8$. (4 pts)

(c) For $n = 16$ forecasters and a known horizon $T = 10{,}000$, find the learning rate η^* that minimizes the bound in (b), the resulting bound on regret, and the implied bound on average regret per round. (4 pts)

(d) *Follow the Leader* (FTL) deterministically follows the forecaster with the smallest cumulative loss so far (ties to forecaster 1). With two forecasters, construct a loss sequence in $[0, 1]$ on which FTL's total loss is about T while the best forecaster's total loss is about $T/2$. Why does Hedge not suffer the same fate? (4 pts)

Problem 3: Fictitious play in matching pennies (20 points)

Row (the "Matcher") and Column (the "Mismatcher") each choose H or T. Row gets $+1$ if the actions match and -1 otherwise; Column gets the negative of Row's payoff.

Both players use *fictitious play*: in each round, each player best-responds to the empirical distribution of the opponent's past actions. The counts are initialized with one fictitious observation: Row starts believing Column has played H once (H:1, T:0), and Column starts believing Row has played T once (H:0, T:1). Both players choose simultaneously, then both counts are updated. **Ties in expected payoff are broken in favor of H.**

Round	Row's counts of Column (H, T)	Column's counts of Row (H, T)	Row plays	Column plays
1	$(1, 0)$	$(0, 1)$	H	H
2				
3				
4				
5				
6				
7				
8				

(a) Complete the table for rounds 2–8 (the counts are those held *before* play in that round). Describe the pattern of play you observe. (8 pts)

(b) What do the empirical frequencies of play converge to as the number of rounds grows? Cite the relevant theorem and state the long-run average payoff. (4 pts)

(c) Prove the following general fact: in any finite two-player game, if fictitious play settles on a fixed pure profile (a^*, b^*) (the same actions are played in every round from some round on), then (a^*, b^*) is a pure Nash equilibrium. (4 pts)

(d) In this example, do the *actions actually played* converge? Explain the difference between convergence of play and convergence of empirical frequencies. (4 pts)

Problem 4: Sharing two GPU clusters (20 points)

Four analysts each submit one job to one of two GPU clusters. If k jobs run on cluster A, each of them takes $c_A(k) = k$ hours; if k jobs run on cluster B, each takes $c_B(k) = k + 1$ hours (cluster B has a one-hour queue overhead). Each analyst chooses a cluster to minimize her own job's completion time.

(a) Write down Rosenthal's potential Φ for this congestion game as a function of (n_A, n_B), and verify that a unilateral switch changes the switching analyst's cost by exactly the change in Φ. (4 pts)

(b) Find all pure Nash equilibria. How many strategy profiles (assignments of named analysts to clusters) are equilibria? (6 pts)

(c) Start from the profile in which all four analysts use B. Run best-response dynamics in which, at each step, one analyst who can strictly reduce her cost switches. Report the sequence of (n_A, n_B) and the values of Φ along the path, and say where the dynamics stop. (4 pts)

(d) The social cost is the total completion time of the four jobs. Find the socially optimal assignment, and compute the price of anarchy and the price of stability. (6 pts)

Problem 5: True or false? ($5 \times 4 = 20$ points)

Answer T or F and give a justification of two to four sentences (a proof sketch, a counterexample, or a precise reference to a result from lecture). The letter alone is worth 1 point.

5.1 In nonatomic selfish routing with affine latency functions, adding a new edge to a network can increase the equilibrium total travel time, but by at most a factor of $4/3$. **T / F**

5.2 In a repeated two-player zero-sum game, if both players use algorithms whose average regret after T rounds is at most ε_T, then the pair of time-averaged mixed strategies is a $2\varepsilon_T$-approximate Nash equilibrium. **T / F**

5.3 If every player in a repeated finite general-sum game uses a no-regret algorithm, then the empirical distribution of joint play converges to the set of Nash equilibria. **T / F**

5.4 Every correlated equilibrium is a coarse correlated equilibrium. **T / F**

5.5 A polynomial-time algorithm for computing a Nash equilibrium of a general two-player finite game (given as payoff matrices) is known. **T / F**

End of Problem Set Exam. Remember to sign the honor statement on the submission page.